The climate change impact of surgical and non-surgical skin cancer appointments: a protocol for a process-based life cycle assessment

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Goal of the study

The goal of this study is to estimate the climate change impact (in kg CO_2-e) of different scenarios in which a patient attends a clinical consultation at a dermatology clinic for the diagnosis or management of skin cancer. We do not intend to make any comparative assessments. We are preparing this life cycle assessment (LCA) for an academic healthcare audience, including dermatologists, health services researchers, and health economists, to inform research and decisions regarding health system sustainability. The audience will not likely be familiar with LCA methodology. The work will be disclosed to the public and we will publish the work in an open access peer-reviewed health journal. We will conduct this LCA in accordance with International Standards ISO 14040:2006 and ISO 14044:2006.[1,2]

Study setting

This study will be conducted based on information from two skin-cancer focused subspecialty dermatology clinics: Sydney Melanoma Diagnostic Centre (SMDC) and Melanoma Dermatology (MD). SMDC private dermatology clinic based at Royal Prince Alfred Hospital in Camperdown, Sydney, New South Wales. MD is a privately operated dermatology clinic part of the Melanoma Institute Australia within the Poche Centre in Wollstonecraft, Sydney, New South Wales.

Functional units and reference flows

In this study, we will investigate the climate change impact of different scenarios in which patients engage with a dermatology clinic for the treatment and/or management of confirmed or suspected melanoma or other skin other cancer. These scenarios can be broadly grouped into two categories:

(1) non-surgical appointments for the identification and monitoring of skin to diagnose lesions suspicious for melanoma or other skin cancers, or
(2) surgical appointments for the excision of skin lesions suspected to be melanoma or other skin cancers.

Clinical consultations for the purpose of monitoring skin lesions typically rely on in-person history, naked-eye physical examination and dermoscopy, a non-invasive technique using a dermatoscope to visualise features of the skin not visible to the naked eye (i.e., a full skin examination). During these kinds of appointments patients may also undergo sequential dermoscopic digital imaging (SDDI) if any suspicious lesions are identified and immediate excision is not required.[3]

Another type of clinical consultation for the purpose of monitoring skin lesions is total body photography (TBP). TBP is a technique where images of the entire body are taken, usually by a melanographer, to act as a baseline and to detect new or macroscopically changing skin lesions. Comparisons to these images can then be made during full skin examinations. TBP may take place during the same visit to the clinic as a full skin examination, or during a dedicated appointment.

Clinical consultations for the purpose of excising lesions suspicious for melanoma or other skin cancers are often arranged following non-surgical skin monitoring appointments. For lesions suspicious for melanoma, excisional biopsies are recommended. Shave and/or punch biopsies may also be conducted if clinical suspicion for melanoma is low, or for lesions suspected to be non-melanoma skin cancer.[6] These appointments may be undertaken at the same, or on a separate visit to the clinic.

In this study each scenario will be considered its own functional unit and no comparisons between reference flows will be made. Details are provided in Table 1.

Table 1. Summary of functional units and reference flows included in this study.

Category	Functional unit
Non-surgical appointments for the monitoring and identification of lesions suspicious for melanoma or other skin cancers	(1) One in-person visit for a dermoscopy guided full skin examination (2) One in-person visit for total body photography (3) One in-person visit for dermoscopy guided full skin examination and total body photography
Surgical appointments for the excision of skin lesions suspicious for melanoma or other skin cancers	(4) One in-person visit for the excision of one lesion suspicious for skin cancer (5) One in-person visit for a shave biopsy of one lesion suspicious for skin cancer (6) One in-person visit for a punch biopsy of one lesion suspicious for skin cancer We may also investigate scenarios where multiple excisions take place during the same appointment, as appropriate.

Data sources

Data will be collected directly using custom data collection sheets developed in consultation with dermatologists and staff at the participating clinics. These will aim to record the number and types of items used during the different types of dermatology appointments. SMDC and MD are recruiting clinics for the MEL-SELF trial (described in detail elsewhere[5]), and additional data are available from the trial. The MEL-SELF trial was approved by Sydney Local Health District (RPAH Zone) Ethics Committee (2019/ETH0531). Publicly available data will also be used as necessary.

Life cycle analysis will be modelled using SimaPro Analyst Release 9.6.0.1 (Pre Sustainability, Amersfoot, the Netherlands). The AusLCI Version 2 (Lifecycles, Melbourne, Australia) and Ecoinvent Version 3.1 (Ecoinvent, Zurich, Switzerland) databases will be used to build the model, with preference given to Australian unit processes from the AusLCI database where possible.

System boundary

For the functional units identified, we will include the following as appropriate:

- The manufacture, use, and disposal of all single-use medical equipment
- The manufacture, use, and sterilisation of all reusable medical equipment
- Patient transport required for in-person appointments
- Energy use for duration of appointment, including for imaging (total body photography) machines
- Energy use for transmission of imaging via teledermatology

We will exclude:

- Staff travel to the skin cancer clinic
- The manufacture and maintenance of clinic electricity infrastructure
- Heating and cooling systems of the clinic
- Healthcare resources that are used outside of the skin cancer clinic, including pathologist review of an excised lesion

System diagrams will be developed as appropriate for the final manuscript.

Data collection and requirements

Medical equipment

Data collection sheets will be developed in consultation with clinic staff and/or by conducting an audit of each clinic's storeroom and supply ordering invoices. Medical equipment will be investigated individually to determine composition, weights, and manufacture origin. The total number of each item used will be estimated from up to ten appointments of each type. Disposable medical equipment will likely include: gauze, drapes, gloves, syringes, needles, biopsy instruments, dressings, alcohol wipes, markers, and paper bed sheets.

Reusable medical equipment is often provided as 'kits' based on the type of excision being performed and are likely to contain a stainless-steel tray, scalpel handle, scissors, needle holders, and forceps. A kit will be obtained, and items will be investigated individually to determine composition and weight. The cleaning and sterilisation procedure will be investigated and included. We will determine and assume an average number of uses per item in consultation with clinic staff and procedure logs.

For all items, the country of manufacture will be determined and the transport (i.e., shipping) of the item to Sydney will be estimated and included in the model.

Patient travel

Patient travel from their home to the clinic will be estimated using data available from the MEL-SELF trial. This includes information about the mode of transport, and distance travelled for each appointment made during the study period. The model will be constructed to include different modes of transport (i.e., car, bus, train, bike, and walk).

Electricity use

Energy consumed by any total body photography imaging machines will be measured using a power meter and/or power meter and logger over a convenient sampling period. Power from computers, monitors, dermatoscopes, and other general equipment used during appointments will be estimated using published data.

Time

Data collection sheets will collect information about the start and end time of each type of appointment.

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